

SOFTWARE REQUIREMENTS SPECIFICATION

Agentic AI Procurement Agent on Beckn Protocol

AI-driven enterprise procurement over an open commerce network

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Prepared for academic project review. The document defines the proposed requirements, architecture, interfaces, constraints, quality goals, and acceptance criteria for the working prototype.

Document Control

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Approval / Review

Role	Name	Status
Team Leader	K. Durga Eswar	Prepared
Faculty Guide	Dr. Ch Laxmi Narayana	To be reviewed
HOD	Dr. G Srinivas	To be reviewed
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Note: Replace all bracketed placeholders with the official team and guide details before submission.

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional, non-functional, integration, security, data, and operational requirements for an Agentic AI Procurement Agent operating over the Beckn open commerce protocol. The system is intended to transform natural-language enterprise procurement requests into structured intent, discover supplier offers, evaluate alternatives, negotiate within organizational policies, obtain approval where required, place an order, track fulfillment, and preserve an auditable record of actions.

1.2 Product Vision

Vision: enable enterprises to procure goods through an AI agent rather than manually searching individual marketplaces. Beckn provides the interoperable commerce communication layer; the AI agent provides planning and decision support; the policy engine provides organizational controls; and the dashboard provides human oversight.

1.3 Intended Audience

- Project review panel and faculty guide.
- Project team members responsible for AI, Beckn integration, backend, and enterprise UX.
- Developers and testers implementing the prototype.
- Future maintainers or evaluators of the system.

1.4 Definitions and References

The project brief describes the classic Beckn transaction sequence using search/select/init/confirm/status. The current official Beckn repository identifies Beckn Protocol v2.0.0 as the LTS line and uses action-specific endpoints including discover, select, init, confirm, status, track, update and cancel. The implementation team shall confirm the exact sandbox/specification baseline before final integration. ■cite■turn0search0■turn0search2■

1.5 Document Conventions

- **SHALL** indicates a mandatory requirement for the prototype.
- **SHOULD** indicates a recommended requirement that may be deferred when justified.
- **MAY** indicates an optional enhancement.
- FR = Functional Requirement; NFR = Non-Functional Requirement; UC = Use Case.

2. Overall Description

2.1 Product Perspective

The proposed system is an enterprise-facing buyer application with an AI agent layer. It communicates with Beckn-compatible provider applications through a protocol adapter. The architecture separates AI decision-making from transaction execution so that every externally significant action is validated by deterministic backend services.

2.2 High-Level Context

User → Procurement UI → Backend API → Agent Orchestrator → Policy Engine → Beckn Adapter → Open Commerce Network → Provider Nodes → Order/Fulfillment → Audit & Memory.

2.3 Major Capabilities

- Natural-language procurement request capture.
- Intent extraction into structured fields such as item, quantity, location, budget, deadline, quality and constraints.
- Supplier and offer discovery through Beckn-compatible APIs.
- Offer normalization, filtering and multi-factor ranking.
- Configurable negotiation with bounded rounds and target prices.
- Policy-based approval routing and human-in-the-loop controls.
- Order confirmation and status tracking.
- Complete agent action and transaction audit trail.
- Organizational procurement memory and supplier-performance history.
- Enterprise dashboard for requests, offers, approvals, orders and audit events.

2.4 Assumptions

- The prototype will have access to a Beckn/ONDC sandbox or a controlled Beckn-compatible mock network.
- At least three provider nodes will be available for deterministic demonstration.
- A suitable LLM/API will be available for natural-language understanding and agent orchestration.
- The project will use synthetic or non-sensitive procurement data during development.
- Human approval remains mandatory for purchases above configured thresholds.
- Network-specific onboarding, authentication and registry details may vary and will be finalized after the sandbox baseline is confirmed.

2.5 Out of Scope for Phase 1

- Unrestricted autonomous purchasing with no approval controls.
- Production-grade financial settlement or banking integration.
- Enterprise-wide ERP replacement.
- Guaranteed real-world supplier fulfillment outside the selected sandbox/network.
- Training a foundation LLM from scratch.
- Fully autonomous legal contract signing.

3. Stakeholders and User Classes

3. Stakeholders and User Classes

Stakeholder / Role	Responsibilities / Needs
Employee / Requester	Submit procurement requests, review status, provide clarifications.
Procurement Manager	Review recommendations, negotiate exceptions, approve purchases.
Finance Approver	Validate high-value transactions, budget and payment terms.
Organization Admin	Configure policies, roles, suppliers and system settings.
Project Team	Develop, test, integrate and maintain the prototype.
Faculty Guide	Review academic progress, scope and technical quality.
Review Panel	Evaluate feasibility, technical understanding and implementation progress.
Supplier / BPP	Publish offers, respond to protocol actions, accept/reject orders and provide status.

3.1 User Roles

The system SHALL implement role-based access control. At minimum, the prototype SHALL support Requester, Approver and Administrator roles. A separate Finance role SHOULD be implemented for high-value purchase workflows.

4. System Scope

4.1 In-Scope End-to-End Lifecycle

Request → Parse → Validate → Discover → Normalize → Rank → Negotiate → Policy Check → Approval → Confirm Order → Track → Audit → Learn.

4.2 Scope Boundaries

Area	Phase-1 Scope	Future Extension
Catalog discovery	Yes	Multi-category catalog intelligence
Offer comparison	Yes	Advanced predictive scoring
Negotiation	Bounded prototype	Multi-round adaptive negotiation
Approval	Yes	Complex enterprise delegation
Payment	Simulated / protocol-level	Real payment rails
Fulfillment	Status tracking	Advanced logistics optimization
Memory	Historical procurement records	Cross-organization learning with privacy controls
Analytics	Basic procurement metrics	Spend intelligence and forecasting

5. Functional Requirements

ID	Requirement	Description
FR-01	User Authentication	The system SHALL authenticate users and associate every procurement request with an organization and user identity.
FR-02	Create Procurement Request	The system SHALL allow a requester to enter a procurement requirement in natural language.
FR-03	Intent Extraction	The system SHALL extract item, quantity, budget, location, deadline and explicit constraints into structured intent.
FR-04	Intent Validation	The system SHALL detect missing or ambiguous mandatory fields and request clarification before external purchasing actions.
FR-05	Search / Discovery	The system SHALL submit a valid discovery request through the configured Beckn adapter.
FR-06	Receive Supplier Results	The system SHALL accept asynchronous supplier/provider responses and associate them with the originating transaction.
FR-07	Offer Normalization	The system SHALL normalize price, quantity, delivery, warranty, seller information and terms into a common internal representation.
FR-08	Offer Filtering	The system SHALL filter offers that violate explicit requirements or configured hard constraints.
FR-09	Offer Ranking	The system SHALL rank eligible offers using configurable scoring criteria.
FR-10	Recommendation	The system SHALL present a recommended offer and a concise explanation of the factors influencing the recommendation.
FR-11	Negotiation	The system SHALL support bounded negotiation using configured target price, limits and maximum rounds.
FR-12	Policy Validation	The system SHALL validate every purchase action against organizational procurement policies.
FR-13	Approval Routing	The system SHALL route purchases to the appropriate approver based on value, category and policy.
FR-14	Order Confirmation	After required approval, the system SHALL initiate order confirmation through the configured Beckn flow.
FR-15	Order Status	The system SHALL retrieve and display order/fulfillment status.
FR-16	Audit Logging	The system SHALL log user requests, agent actions, policy decisions, approvals, protocol events and order outcomes.
FR-17	Memory	The system SHOULD store historical procurement outcomes, supplier performance and organizational preferences.
FR-18	Notifications	The system SHOULD notify users when clarification, approval or significant order status changes require attention.

6. Agentic AI Requirements

6.1 Agent Responsibilities

The AI layer SHALL be implemented as a controlled agent workflow rather than a free-form chatbot. The agent may reason over user intent and tool results, but deterministic backend services SHALL enforce authorization, policy and transaction safety.

6.2 Agent Components

Component	Purpose	Key Inputs	Outputs
Intent Agent	Convert natural language into structured intent	User request	Intent JSON
Discovery Agent	Plan and execute supplier discovery	Intent + policies	Supplier offers
Evaluation Agent	Compare eligible offers	Offers + weights	Ranked offers
Negotiation Agent	Generate bounded counter-offers	Offer + negotiation policy	Counter-offer / decision
Approval Agent	Determine approval path	Purchase + policy	Approval request / auto-pass
Explanation Agent	Explain recommendation and actions	Decision trace	Human-readable explanation

6.3 Agent Tool Contract

The agent SHALL interact with backend capabilities through explicit tools such as `search_suppliers()`, `get_offer_details()`, `score_offers()`, `negotiate()`, `check_policy()`, `request_approval()`, `confirm_order()`, and `get_status()`.

6.4 Guardrails

- The LLM SHALL NOT directly execute database writes or external purchase operations.
- Tool calls SHALL be validated by the backend.
- Seller/provider content SHALL be treated as untrusted data and SHALL NOT override system or organization policy.
- Negotiation SHALL stop when the configured round, price or policy limits are reached.
- High-value transactions SHALL require explicit human approval.
- Agent actions SHALL be traceable to the originating request and tool invocation.

7. Beckn / ONDC Integration Requirements

7.1 Protocol Baseline

The current official Beckn Protocol v2.0.0 repository describes a stable transport contract with action-specific APIs, asynchronous callbacks, and coverage for discovery, ordering, fulfillment and post-fulfillment. The authoritative API artifact is the OpenAPI 3.1.1 specification. See the official Beckn v2.0.0 LTS specification in the references.

7.2 Prototype Integration Flow

Buyer application / BAP → Beckn Adapter → Network / Discovery → Provider/BPP → callback → Offer normalization → Agent evaluation.

7.3 Required Protocol Capabilities

- Discovery of relevant providers/catalog items.
- Selection of one or more items/offers.
- Initialization of the commercial transaction.
- Confirmation of the order after approval.
- Retrieval of order status and, where supported, tracking.
- Handling of acknowledgements and asynchronous callbacks.
- Request/response signing and verification as required by the selected Beckn version.
- Standardized error handling and transaction correlation.

7.4 Version Compatibility Rule

The project brief references the classic /search, /on_search, /select, /on_select, /init, /on_init, /confirm, /on_confirm and /status style. The implementation team SHALL document the exact protocol version used in the prototype and map the brief's terminology to that version. Beckn v2.0.0 LTS currently uses /discover and /on_discover for discovery and retains select/init/confirm/status families.

7.5 Mock Provider Requirement

For deterministic demonstration, the prototype SHOULD include at least three mock provider nodes implementing the agreed protocol contract. Each provider SHALL expose different price, delivery, rating and warranty characteristics so that the agent has a meaningful decision to make.

8. Enterprise Policy, Approval and Audit

8.1 Policy Engine

Policies SHALL be represented as structured configuration rather than being embedded only in prompts. Example policies include maximum spend, minimum supplier rating, permitted categories, negotiation limits, payment terms and approval thresholds.

8.2 Example Approval Matrix

Purchase Value	Required Action
Up to ■50,000	Automatic execution if all policies pass
■50,001 – ■5,00,000	Manager approval
■5,00,001 – ■25,00,000	Finance approval
> ■25,00,000	Senior management approval

8.3 Audit Trail

- User request and parsed intent.
- All discovered offers and filtering decisions.
- Scoring inputs and recommendation.
- Negotiation messages and policy outcomes.
- Approval request, approver identity and decision time.
- Protocol actions, acknowledgements, callbacks and order identifiers.
- Order status changes and post-purchase feedback.

8.4 Explainability

For each recommendation, the system SHALL show the principal decision factors and relevant constraints. The explanation SHALL distinguish factual supplier data from AI-generated interpretation.

9. Data Requirements

9.1 Core Entities

Entity	Important Attributes
User	id, organization_id, role, name, email, status
Organization	id, name, policies, approval configuration
ProcurementRequest	id, requester, raw_text, structured_intent, status, timestamps
Supplier	id, provider_id, name, rating, capabilities
Offer	id, request_id, supplier_id, item, quantity, unit_price, delivery, warranty, terms, score
Negotiation	id, offer_id, rounds, messages, target, final_offer, outcome
Approval	id, request_id, approver, threshold, decision, timestamp
Order	id, request_id, supplier_id, total, protocol_ids, status
AgentAction	id, request_id, agent, tool, input, output, result, timestamp
AuditEvent	id, actor, action, object, metadata, timestamp
MemoryRecord	id, organization_id, type, content, embedding, source

9.2 Data Retention

Audit and transaction records SHALL be retained for the duration required by the project or institutional policy. Prototype data SHALL avoid real financial credentials, personal secrets and unnecessary personally identifiable information.

10. External Interfaces

10.1 User Interface

- Procurement request entry.
- Agent activity timeline.
- Supplier/offer comparison.
- AI recommendation and explanation.
- Negotiation history.
- Approval queue.
- Order and fulfillment tracking.
- Audit trail viewer.
- Policy configuration for administrators.

10.2 Backend API

The backend SHOULD expose REST APIs for authentication, requests, offers, approvals, orders, audit events and dashboard data. Long-running agent operations SHOULD be asynchronous and expose request status.

10.3 Beckn Interface

The Beckn adapter SHALL isolate protocol-specific message construction, signing, callback handling and version-specific mappings from the rest of the application. This prevents the AI layer from depending directly on protocol payload details.

10.4 LLM Interface

The LLM integration SHALL support structured outputs/tool calls where available. Prompts SHALL include only the context necessary for the task, and sensitive secrets SHALL never be passed to the model.

10.5 Example Internal API Contracts

Endpoint	Purpose
POST /api/procurement/requests	Create procurement request
POST /api/procurement/{id}/run	Start agent workflow
GET /api/procurement/{id}	Retrieve request and current state
GET /api/procurement/{id}/offers	Retrieve normalized offers
POST /api/procurement/{id}/negotiate	Start / continue bounded negotiation
POST /api/procurement/{id}/approve	Approve or reject purchase
POST /api/orders/{id}/confirm	Confirm approved order
GET /api/orders/{id}/status	Retrieve order status
GET /api/audit/{request_id}	Retrieve audit trail

11. Non-Functional Requirements

ID	Quality Area	Requirement
NFR-01	Performance	The UI should acknowledge user actions within 2 seconds under normal prototype load. Long-running network/agent tasks shall be asynchronous.
NFR-02	Reliability	Protocol failures shall not silently create duplicate orders. Transaction state shall be persisted before critical transitions.
NFR-03	Scalability	The backend shall be designed so additional providers and procurement categories can be added without rewriting the agent.
NFR-04	Maintainability	Protocol integration, AI logic, policy logic and UI shall be separated into modular components.
NFR-05	Usability	A first-time requester should be able to submit a basic procurement request without protocol knowledge.
NFR-06	Explainability	Every recommendation shall expose key decision factors and relevant constraints.
NFR-07	Observability	Logs, metrics and trace identifiers shall allow a failed procurement workflow to be diagnosed.
NFR-08	Security	Authentication, authorization, secret management, signed protocol messages and input validation shall be implemented.
NFR-09	Auditability	Material actions shall have immutable or append-only audit records for the prototype.
NFR-10	Interoperability	The Beckn adapter shall conform to the selected protocol version and validate payloads against the selected schemas.
NFR-11	Testability	Core agent, policy, protocol mapping and approval components shall have automated tests.
NFR-12	Portability	The application shall run locally through Docker and be deployable to a standard cloud environment.

12. Security and Privacy

12.1 Authentication and Authorization

- Use OAuth 2.0 / OIDC or a secure JWT-based mechanism for prototype authentication.
- Implement role-based access control for Requester, Approver, Finance and Administrator roles.
- Approvers SHALL only approve requests within their authorization scope.

12.2 Secret Management

- API keys and protocol credentials SHALL be stored in environment secrets or a secret manager.
- Secrets SHALL never be committed to source control or displayed in logs.
- LLM keys SHALL remain server-side.

12.3 Agent Security

- External supplier text SHALL be treated as untrusted.
- Tool arguments SHALL be schema-validated.
- The agent SHALL not be allowed to alter policies.
- The agent SHALL not bypass approval thresholds.
- Prompt-injection attempts in supplier responses SHALL be ignored as instructions.

12.4 Beckn Security

The selected Beckn specification requires its defined request/response authentication and signature semantics. The adapter SHALL implement the signing and verification mechanism required by the selected network/version.

12.5 Privacy

Only data required for procurement shall be collected. The prototype SHOULD use synthetic user and supplier data wherever possible. Personal or financial secrets shall not be stored in LLM prompts, embeddings or demo logs.

13. System Architecture

13.1 Logical Architecture

The recommended architecture is a modular monolith for the first prototype, with clear internal boundaries. Microservices may be introduced later if scale or deployment requirements justify them.

Layer	Components
Presentation	Next.js, TypeScript, Tailwind/shadcn UI
Application API	FastAPI, authentication, request/approval/order APIs
Agent Orchestration	LangGraph or equivalent stateful workflow engine
AI Services	LLM provider, structured output, embeddings
Domain Services	Intent, evaluation, negotiation, policy, approval
Protocol Adapter	Beckn version-specific message mapping, signing, callbacks
Data	PostgreSQL, pgvector, Redis
Observability	Structured logs, trace IDs, metrics
Infrastructure	Docker, CI/CD, cloud deployment

13.2 Architecture Principle

LLM proposes; deterministic services validate; policy controls; human approves when required; protocol adapter executes; audit records everything.

13.3 Suggested Repository Structure

```
backend/  
■■■ api/  
■■■ agents/  
■■■ domain/  
■■■ policy/  
■■■ beckn/  
■■■ models/  
■■■ services/  
■■■ audit/  
■■■ tests/  
  
frontend/  
■■■ app/  
■■■ components/  
■■■ lib/  
■■■ tests/
```

14. Use Cases and Workflows

ID	Use Case	Actor	Trigger	Expected Result
UC-01	Submit Procurement Request	Requester	User submits a natural-language requirement.	Structured request created.
UC-02	Clarify Requirement	Agent + Requester	Mandatory information is missing or ambiguous.	Request becomes complete and valid.
UC-03	Discover Offers	Agent	Valid request is available.	Supplier offers collected.
UC-04	Compare Offers	Agent	Multiple offers are available.	Ranked shortlist and explanation.
UC-05	Negotiate	Negotiation Agent	Eligible offer can be improved within policy.	Accepted counter-offer or negotiation termination.
UC-06	Approve Purchase	Manager / Finance	Policy requires human approval.	Approved or rejected purchase.
UC-07	Confirm Order	System	Approval completed.	Order confirmation sent.
UC-08	Track Order	Requester	Order exists.	Current status displayed.
UC-09	Review Audit Trail	Approver / Admin	Procurement request exists.	Traceable action history.

14.1 Primary Workflow

1. User enters request → 2. Intent agent extracts fields → 3. Validator checks completeness → 4. Discovery agent searches providers → 5. Callback results are normalized → 6. Evaluation agent ranks offers → 7. Negotiation agent attempts bounded negotiation → 8. Policy engine checks transaction → 9. Approval workflow runs if needed → 10. Beckn adapter confirms order → 11. Status is tracked → 12. Audit and memory are updated.

14.2 Exception Workflows

- No offers found → broaden search criteria or ask user for acceptable alternatives.
- All offers violate budget → request revised budget or user confirmation.
- Supplier negotiation fails → retain best valid offer or return shortlist to user.
- Policy violation → block action and show the violated rule.
- Approval rejected → terminate order action and retain audit trail.
- Protocol timeout → retry safely according to configured retry policy; do not duplicate critical transactions.
- Invalid provider response → quarantine response, log error, and continue with other providers where possible.

15. Testing and Acceptance Criteria

15.1 Testing Strategy

- Unit tests for intent parsing adapters, scoring, policy rules and data validation.
- Integration tests for Beckn request/callback mapping and provider mocks.
- Agent workflow tests using deterministic fixtures.
- End-to-end tests covering request → discovery → approval → order.
- Security tests for RBAC, secret exposure and prompt-injection scenarios.
- UI tests for request creation, offer review and approval.

15.2 Phase-1 Acceptance Criteria

ID	Acceptance Criterion
AC-01	A user can submit a procurement request in plain English.
AC-02	The system produces structured intent with item, quantity, budget, location and deadline where provided.
AC-03	The system can discover offers from at least three configured provider nodes or the approved sandbox.
AC-04	At least five meaningful offer attributes can be compared.
AC-05	The system produces a ranked recommendation with an explanation.
AC-06	The agent can execute at least one bounded negotiation scenario.
AC-07	A configured policy can force human approval above a threshold.
AC-08	An approver can approve/reject a transaction from the dashboard.
AC-09	An approved order can progress through the selected Beckn transaction flow.
AC-10	The order status can be displayed after confirmation.
AC-11	The complete action history is visible in an audit trail.
AC-12	The prototype can be demonstrated end-to-end using repeatable test data.

15.3 Definition of Done

- Feature is implemented and code-reviewed.
- Automated tests pass.
- No critical security issue is known.
- Logs and error handling are present.
- Feature is demonstrated with representative data.
- Relevant SRS/API documentation is updated.

16. Project Constraints, Risks and Mitigations

ID	Risk	Severity	Mitigation
R-01	Protocol version mismatch	High	Confirm the exact Beckn/ONDC baseline before coding; isolate version-specific logic in the adapter.
R-02	Sandbox availability / onboarding	High	Build deterministic mock BPPs in parallel with sandbox integration.
R-03	LLM hallucination	High	Use structured outputs, tool constraints and deterministic validation.
R-04	Unsafe autonomous purchase	High	Policy engine + approval thresholds + backend authorization.
R-05	Provider response inconsistency	Medium	Schema validation, normalization and error quarantine.
R-06	Scope creep	Medium	Prioritize end-to-end MVP before advanced analytics.
R-07	Integration delays	Medium	Define API contracts early and test mock network first.
R-08	Demo instability	Medium	Use seeded test data, deterministic mock providers and fallback demo mode.

16.1 Key Constraints

- Three-person student project team.
- Limited project timeline and review milestones.
- External network/sandbox dependencies.
- Need to demonstrate protocol behavior without depending entirely on live commercial infrastructure.
- LLM output variability must be controlled for repeatable evaluation.

17. Implementation Roadmap

Phase	Focus	Deliverables
Phase 1	Foundation	Requirements, SRS, architecture, protocol study, API contracts
Phase 2	Beckn Core	BAP/adapter, mock BPPs, discovery and transaction flow
Phase 3	Agent Core	Intent, orchestration, evaluation and explanation
Phase 4	Negotiation	Negotiation strategy, policy limits, counter-offers
Phase 5	Enterprise Controls	Approval, RBAC, audit, memory
Phase 6	Integration	End-to-end flow, sandbox validation, failure handling
Phase 7	Quality	Testing, security, observability, performance
Phase 8	Final Demo	Deployment, documentation, presentation and live demonstration

17.1 Team Work Allocation

Team Member	Primary Responsibility	Secondary
Intern 1 / Beckn Integration	Protocol adapter, BAP/BPP integration, sandbox	Backend APIs, protocol testing
Intern 2 / AI Agent	Intent, orchestration, evaluation, negotiation, memory	AI testing, prompts/tool schemas
Intern 3 / Enterprise UX	Dashboard, approval workflows, audit, policies	UX testing, documentation

17.2 Milestone Principle

Every milestone should produce an executable increment. The first technical milestone is not a polished UI; it is a successful buyer-to-provider protocol interaction. The first AI milestone is not a chatbot; it is a structured intent that can drive a real tool call. The final milestone is a repeatable end-to-end procurement demo.

18. Glossary

Term	Definition
Agentic AI	An AI system that can plan, use tools, observe results and take bounded actions toward a goal.
BAP	Buyer-side application/platform participating in a Beckn network.
BPP	Provider-side application/platform exposing supply or services to the network.
Beckn Protocol	Open protocol for interoperable value exchange across independently operated network participants.
ONDC	Open Network for Digital Commerce, an open-network ecosystem using Beckn-derived protocol specifications.
Discovery	Process of finding relevant providers/catalog items from a network.
Intent	Structured representation of what a consumer wants to procure.
Offer	A supplier-provided commercial option including item, price, fulfillment and terms.
Policy Engine	Deterministic component that evaluates organizational rules and authorization.
Human-in-the-loop	Design in which a human must review or approve selected actions.
Audit Trail	Chronological record of actions, decisions, approvals and protocol events.
Provider Node	Supplier-side participant capable of responding to protocol interactions.
Sandbox	Controlled environment for development and testing.

19. References

- [1] Beckn Protocol v2.0.0 LTS — official protocol specification repository. Defines the current action-specific transport contract, asynchronous callback model and protocol API surface.
- [2] Beckn Protocol v2.0.0 OpenAPI specification — authoritative API definition.
- [3] Beckn Protocol Core Specification repository — background on Beckn as an open protocol and the classic transaction model.
- [4] Beckn Core Schemas — domain-agnostic schema definitions for Beckn Protocol v2.
- [5] ONDC Official Protocol Specifications — ONDC core protocol specifications and Beckn-derived APIs.

Reference URLs

<https://github.com/beckn/protocol-specifications-v2>
<https://github.com/beckn/protocol-specifications-v2/blob/main/api/v2.0.0/beckn.yaml>
<https://github.com/beckn/protocol-specifications>
<https://github.com/beckn/schemas>
<https://github.com/ONDC-Official/ONDC-Protocol-Specs>

Implementation note: The project team shall treat the exact sandbox documentation and selected network specification as the authoritative integration contract at implementation time. The SRS intentionally separates protocol-specific details behind a Beckn adapter so that the rest of the system remains stable if the network version changes.

End of Software Requirements Specification